

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
COMPUTER NETWORKS & INTERNET of THINGS (18B11CS311)
BTECH 6TH SEM Even_2021
TUTORIAL-10 & 11 (Week 10 & 11) (19 April – 1 May 2021)
DATALINK LAYER

Q1. (i) Using ODD Parity, add the correct parity bit to the end of the following bit sequences:

- a. 111010001
- b. 00101
- c. 1001
- d. 0
- e. 11111
- f. 000000

(ii) Using EVEN Parity, add the correct parity bit to the end of the following bit sequences:

- a. 000000
- b. 101000
- c. 11
- d. 11011010011
- e. 0110

Q2. Suppose you wish to transmit the bit sequence 110100011. Suppose also that you want to be able to detect errors using a CRC code. Let the CRC polynomial be given as $x^4 + x^2 + 1$. What bit sequence is ultimately transmitted? Suppose that the bit sequence 101001100 is received at a host. If the polynomial is the same one used before, is there an error with this transmission?

Q3. Suppose you want to transmit the message 11001001 and protect it from errors using the CRC polynomial $x^3 + 1$.

- a. Use polynomial long division to determine the message that should be transmitted.
- b. Suppose the left-most bit of the message is inverted. What is the result of the calculation? How does the receiver know that an error has occurred?

Q4. Consider the 7-bit generator, $G=10011$, and suppose that D has the value 1010101010. What is the value of R ?

Q5. The following Hamming coded (single bit correction) string 010101010111 using even parity was received.

1	2	3	4	5	6	7	8	9	10	11	12
0	1	0	1	0	1	0	1	0	1	1	1

Is there an error? If yes, in which position?

Q6. Suppose four active nodes—nodes A, B, C and D—are competing for access to a channel using slotted ALOHA. Assume each node has an infinite number of packets to send. Each node attempts to transmit in each slot with probability p . The first slot is numbered slot 1, the second slot is numbered slot 2, and so on.

- a. What is the probability that node A succeeds for the first time in slot 5?
- b. What is the probability that some node (either A, B, C or D) succeeds in slot 4?
- c. What is the probability that the first success occurs in slot 3?
- d. What is the efficiency of this four-node system?

Q7. A group of N stations share a 56-kbps pure ALOHA channel. Each station outputs a 1000-bit frame on average once every 100 sec, even if the previous one has not yet been sent (e.g., the stations can buffer outgoing frames). What is the maximum value of N ?

Q8. Suppose nodes A and B are on the same 10 Mbps broadcast channel, and the propagation delay between the two nodes is 325 bit times. Suppose CSMA/CD and Ethernet packets are used for this broadcast channel. Suppose node A begins transmitting a frame and, before it finishes, node B begins transmitting a frame. Can A finish transmitting before it detects that B has transmitted? Why or why not? If the answer is yes, then A incorrectly believes that its frame was successfully transmitted without a collision. *Hint:* Suppose at time $t = 0$ bits, A begins transmitting a frame. In the worst case, A transmits a minimum-sized frame of $512 + 64$ bit times. So A would finish transmitting the frame at $t = 512 + 64$ bit times. Thus, the answer is no, if B's signal reaches A before bit time $t = 512 + 64$ bits. In the worst case, when does B's signal reach A?

Q9. Suppose the round-trip propagation delay for Ethernet is $46.4 \mu\text{s}$. This yields a minimum packet size of 512 bits (464 bits corresponding to propagation delay+48 bits of jam signal).

- a. What happens to the minimum packet size if the delay time is held constant, and the signalling rate rises to 100 Mbps?
- b. What are the drawbacks to so large a minimum packet size?
- c. If compatibility were not an issue, how might the specifications be written so as to permit a smaller minimum packet size?

Q10. Ethernet is 1-persistent. What does this mean? How does Ethernet handle congestion?

Q11. Explain the difference between an IP address and a MAC address, and why we cannot use only MAC addresses or IP addresses alone.

Q12. With respect to Ethernet answer the following questions briefly:

- a. What is the use of preamble in Ethernet frame/
- b. Why the minimum size of 64 bytes is defined in Ethernet frame header?
- c. Why the jamming signal is of 48 bit times?
- d. What is the relation between minimum frame size, round trip propagation and collision detection time in Ethernet?